

FORMAL LANGUAGES AND AUTOMATA THEORY

QUIZ # 2 - SOLUTION

REG#: _____

NAME: _____

COURSE CODE: CS224

INSTRUCTOR: MUHAMMAD SAJID ALI

TOTAL MARKS: 12

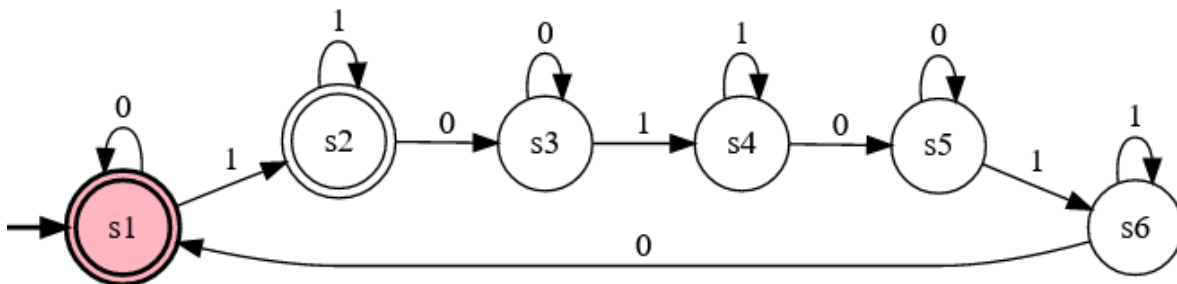
TOTAL TIME: 20 mins

- 1) Consider the language L defined over alphabet $\Sigma = \{1, 0\}$ in which the total number of substrings 10 's is divisible by 3 no matter how they are distributed in a word: **(07 MARKS)**

A) Write a regular expression for the given language. **(2 MARKS)**

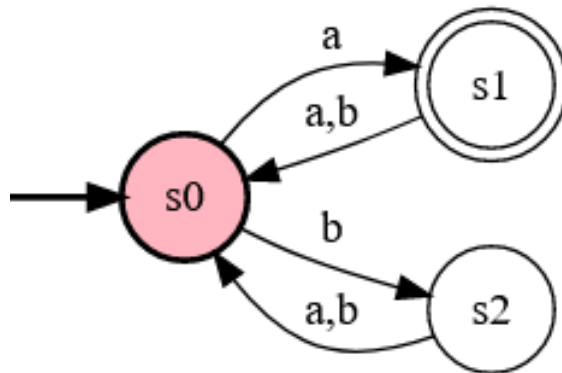
$$0^* + 1^* + (0^* 1^+ 0^+ 1^+ 0^+ 1^+ 0^+)^+$$

B) Build a Finite State Machine (FSM) for the given language. **(5 MARKS)**



2) Describe in English the language accepted by the following DFA

(2.5 MARKS)



$L = \{\text{DFA accepts all words of odd length that ends with a}\}$

3) Write regular expression for the language accepted by the above DFA

(2.5 MARKS)

R.E = $(aa + ab + ba + bb)^* a$